

Pointer `p` passed to `FUTEX_WAIT` and `FUTEX_WAKE` can be the address of any int-sized piece of data

The OS doesn't need to know about this data before you call `futex`

The `futex` system call doesn't care what meaning you ascribe to the values of `*p`

We store `futex` waiter queues in kernel space, when a thread call `FUTEX_WAIT`, we create new wait queue keyed by the address `p`. When all waiters are woken, queue memory is freed.

Barging: a sleep thread is woken, but another new thread acquire the lock before the woken thread.

Problem: wasteful to call `FUTEX_WAKE` when there are no waiters